SUSHANTH REDDY

Data Engineer

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PROFESSIONAL SUMMARY:

- **6+ years** of professional experience as a Data Engineer, with a proven track record of designing, implementing, and optimizing Azure and AWS based data pipelines.
- Expertise in managing large-scale data processing, integration, and analytics in cloud and on-premises environments.
- Proficient in data engineering technologies such as **Python**, **SQL**, **Apache Spark**, **Kafka**, and **Hadoop**.
- Experienced in designing and maintaining **ETL pipelines** for ingesting, transforming, and loading data from multiple sources into data warehouses and data lakes.
- Adept in **data warehousing technologies** like **Snowflake**, **Synapse Analytics**, and **Redshift** to enable scalable and high-performance analytics solutions.
- Familiar with both **batch** and **real-time data processing** paradigms using tools like **Stream Analytics** and **Kafka**.
- Strong understanding of **cloud platforms** (Azure, AWS, GCP) and experience in building cloud-native data solutions for optimized scalability.
- Skilled in **data governance**, ensuring data quality, consistency, and security across multiple systems.
- Hands-on experience in automating repetitive tasks with **Apache Airflow**, improving workflow efficiency and reducing manual errors.
- Designed and implemented **data pipelines** to process large volumes of **financial**, **healthcare**, and **customer** data in various industries.
- Strong problem-solving abilities in debugging and troubleshooting data-related issues.
- Proficient in creating **data models**, **schemas**, and designing efficient databases for reporting and analysis.
- Focused on delivering high-quality data solutions by working with cross-functional teams such as data scientists, analysts, and business stakeholders.
- Built complex **data visualizations** in tools like **Tableau** and **Power BI** to communicate insights effectively.
- Expertise in **data integration** across various systems, including third-party APIs and **legacy** systems.
- Strong grasp of **machine learning pipelines** for preprocessing data and feeding it to models for predictive analytics.
- Dedicated to enhancing **data accessibility**, ensuring data can be leveraged effectively by end-users for business decision-making.
- Adept at working in **Agile** environments and delivering timely data solutions with an emphasis on continuous improvement.
- Excellent communication skills, capable of interacting with technical and non-technical stakeholders to bridge the gap and ensure project success.

TECHNICAL SKILLS:

Programming Languages: Python, SQL, Java, Scala, Shell Scripting

Big Data Processing Frameworks: Apache Spark, Apache Flink, Apache Hadoop, PySpark

Data Streaming & Real-Time Processing: Apache Kafka, Apache Storm, Azure Stream Analytics

ETL Tools & Data Integration: Apache Airflow, Talend, Informatica, Apache NiFi

Cloud Data Platforms: AWS (Redshift, Glue, S3, EC2, Athena, Lambda), Azure (Data Factory, Synapse Analytics, Databricks, Data Lake, DevOps, Key Vault, Logic Apps), Google Cloud (BigQuery, Dataflow)

Data Warehousing: Snowflake, Amazon Redshift, Google BigQuery, Teradata

Database Management Systems: MySQL, PostgreSQL, MongoDB, SQL Server

Data Lakes & Data Pipelines: Data Lake Architecture, S3-based Data Lakes, Pipeline Automation, ETL Optimization

Data Modeling & Schema Design: Dimensional Modeling, Star Schema, Snowflake Schema, Normalization/De-Normalization

Data Governance & Quality: Data Lineage, Data Validation, Data Quality Checks, Data Security (Encryption, Role-based Access Control)

Version Control & CI/CD: Git, Jenkins, Bitbucket, GitLab, GitHub
Containerization & Orchestration: Docker, Kubernetes, Airflow DAGs, Data Pipeline Automation
Machine Learning Pipelines: Data Preprocessing for ML Models, Scikit-learn, TensorFlow (data feeding)
Batch & Real-Time Data Processing: Batch Jobs, Data Streaming, Event-driven Architectures
Data Visualization & Reporting: Tableau, Power BI, SQL Reporting, Jira, Data Analytics, Data Dashboards
Operating Systems: Linux, Windows, Ubuntu (Data Pipeline Management)

PROFESSIONAL EXPERIENCE:

Cleveland Clinic, *Cleveland, OH* *June 2024 - Present*
Data Engineer

- Involved in the design and implementation of Azure based **data pipelines** to process healthcare data, ensuring **compliance with regulatory standards** such as HIPAA.
- Led the migration of legacy data pipelines to Azure-based solutions, improving system performance and scalability.
- Developed and optimized real-time streaming data pipelines using **Azure Event Hubs, Azure Stream Analytics,** and **Apache Spark**, improving healthcare data processing efficiency.
- Integrated data from various healthcare sources, including **patient records**, **medical devices**, and **clinical databases**.
- Collaborated with data scientists to design data models that power advanced analytics and **predictive models** for patient care.
- Automated routine data processing tasks using **Data Factory** and **Azure Logic Apps**, significantly reducing manual intervention and error rates.
- Designed and implemented a scalable **data lake** architecture using **Azure Data Lake Storage** and **Azure Data Factory**, enabling faster data retrieval and analytics.
- Optimized SQL queries and data extraction processes, reducing **data retrieval times by 30%**.
- Developed a robust **data governance framework** to ensure data quality, integrity, and security in compliance with healthcare standards.
- Provided actionable insights into patient care and hospital operations through data- driven visualizations in **Power BI**.
- Worked closely with data architects to refine the overall data infrastructure for scalability and performance.
- Built data models in **Snowflake** to allow for high-performance querying and reporting.
- Worked with healthcare teams to design reports and dashboards that track critical metrics like patient satisfaction and operational efficiency.
- Conducted regular audits of data pipelines to ensure accuracy and identify potential bottlenecks in the data flow.
- Optimized **ETL processes**, leading to a 40% reduction in processing time for large datasets.
- Implemented an automated data validation process using **Data Factory** and **Python scripts**, ensuring the consistency and accuracy of incoming data from diverse sources.
- Regularly monitored and optimized the performance of data systems utilizing **Azure Monitor** and Spark UI, identifying bottlenecks, reducing downtime and improving overall system reliability.
- Introduced git-based data **version control** in collaboration with the development team, ensuring safe handling of data changes and pipeline updates.
- Contributed to the development of a comprehensive **data pipeline monitoring** dashboard for real-time system performance tracking.

Ally Bank, *Chennai, India* *Dec 2020 - Dec 2022*
Data Engineer

- Built and maintained **ETL pipelines** to process large-scale transactional and financial data for reporting and decision-making.
- Integrated data from internal systems and third-party APIs into a centralized **data warehouse** built on **Snowflake**.
- Designed and developed automated data pipelines using **Apache Airflow**, reducing manual intervention by 50%.
- Implemented real-time data processing pipelines with **Apache Kafka** for timely analysis of financial transactions.
- Developed SQL-based data models and optimized **queries** for improved performance on large financial datasets.
- Implemented data extraction strategies from legacy systems, transforming the data into more usable formats for business.
- Led the migration of financial data infrastructure to **Azure**, improving scalability and reducing operational costs.

- Worked with data scientists to preprocess financial data using Databricks for machine learning models, focusing on fraud detection and risk analysis.
- Developed interactive **dashboards** in **Tableau** for senior management to visualize key performance indicators and financial health.
- Spearheaded the creation of **data warehouse architecture** on **Azure Synapse** to centralize financial and customer data.
- Implemented **data versioning** practices in Azure DevOps to safeguard data consistency and ensure accuracy across different stages of data processing.
- Established **data quality checks** throughout the ETL process, ensuring accuracy and reducing errors in financial reporting.
- Reduced manual reporting efforts by creating **self-service BI tools** for internal stakeholders to generate reports on-demand.
- Led the integration of **external financial datasets**, improving customer insights and enriching business intelligence.
- Introduced **data security protocols**, implementing RBAC and data encryption in the ETL pipelines to ensure financial data was handled securely, in compliance with industry standards.

Equinix, *Chennai, India* *Jan 2018 - Dec 2020*
Data Engineer

- Developed **ETL pipelines** for large-scale data processing, enabling seamless integration of data from various sources across Equinix's global data centers.
- Worked with **Apache NiFi** to ingest and transform operational data, feeding it into a centralized data warehouse for analysis.
- Built and maintained data models in Amazon **Redshift**, optimizing storage and access for performance improvements in analytics.
- Collaborated with cross-functional teams to design custom data solutions for **networking**, **customer**, and **data center** analytics.
- Automated **data processing workflows** using Amazon **Step Functions** and **Airflow**, improving pipeline efficiency and reducing manual effort.
- Designed **data architecture** solutions for large-scale data integration, ensuring high availability and minimal downtime.
- Implemented **data partitioning** strategies, improving the query performance of large datasets stored in **S3** and **Redshift**.
- Worked with the data analytics team to produce business intelligence reports for internal stakeholders, tracking key performance indicators.
- Optimized **ETL processes** to handle increased data volumes, resulting in a 25% performance improvement.
- Led the migration of legacy data systems to a more **cloud-based infrastructure** (AWS), leveraging Amazon **S3**, AWS **Glue**, and Redshift for improved scalability and flexibility.
- Regularly monitored and fine-tuned **data flow processes** using AWS CloudWatch, improving the efficiency of real-time analytics.
- Collaborated with IT and network teams to integrate AWS monitoring systems, ensuring data was available in near real-time.
- Worked on **data security** measures to protect sensitive customer and operational data across Equinix's systems.
- Established **data governance protocols** to ensure proper data management and consistency.

EDUCATION:

Master of Science in Information Systems
University of North Texas, May 2024

CERTIFICATION:

Microsoft Certified: Azure Data Engineer Associate (DP-203)

Microsoft Certified: Fabric Data Engineer Associate (DP-700)